REVISED GROUP PROPOSAL / SOCIAL NETWORK ANALYSIS

Structural Resilience and Hub Dependence in US Airline Networks

A hypothesis-driven comparison of airline connectivity and vulnerability to hypothetical airport closures.

Study design	Three US marketing-carrier networks: American Airlines (AA), Delta Air Lines (DL), Southwest Airlines (WN).
Data window	1 April–30 June 2025 | US Bureau of Transportation Statistics (BTS).
Project team	Four MSc Artificial Intelligence students | Three-week project.

Core design choice	Original descriptive questions on hubs and communities become **explanatory analyses** for resilience and airport criticality. Degree-preserving randomized networks provide the reference against which the observed structure is assessed.

1 Research objective

We investigate how airline network structure relates to resilience when airports are removed. The central aim is to determine whether observed vulnerability can be explained by each airline's **in-degree and out-degree sequences** or whether the arrangement of connections, communities, and bridging positions contributes additional information. We also test which existing network measures best identify structurally critical airports.

2 Research questions and hypotheses

RQ1 | Structural determinants of resilience

To what extent do airline networks' structural properties influence their resilience to random and targeted airport disruptions beyond what is expected from their degree sequences?

- **H1 — Targeted vulnerability:** Removing airports selected by degree or betweenness causes larger connectivity losses than removing the same number at random.
- **H2 — Beyond-degree structure:** Resilience in an observed airline network differs systematically from degree-preserving randomized counterparts under matched removal experiments.

RQ2 | Identifying critical airports

Which network measures best predict the loss of connectivity caused by a single airport removal, and do community-bridging measures add information beyond degree?

- **H3 — Bridging importance:** Airports with stronger inter-community bridging roles tend to cause larger removal losses than airports with similar degrees but weaker bridging roles.

3 Hypothesis-to-analysis framework

This framework is **part of the research design, not an optional final discussion**. Agree on comparisons and outcomes before running the experiments; then reuse the same framework to organize the results section.

Test	Planned comparison	Evidence to report
H1 Targeted vs. random	For the same airline and removal fraction, compare degree-/betweenness-targeted losses against repeated random removals.	Differences in normalized efficiency, reachability, and largest SCC; resilience curves and random-removal intervals.
H2 Observed vs. degree-preserving null	Under identical simulation rules, compare each real network with its own directed degree-preserving rewirings.	Observed outcome, null mean, 2.5–97.5% null interval, and direction of any deviation. A null difference does not isolate its cause.
H3 Added value beyond degree	Compare airports with similar degrees, and compare a degree-only criticality prediction with one adding existing bridging measures.	Actual single-airport efficiency losses; Spearman/top-k results and degree-controlled predictive gain or matched comparisons.

4 Dataset and network definitions

• **Source:** BTS Marketing Carrier On-Time Performance data for 1 April–30 June 2025; use the marketing-carrier perspective consistently, including regional partners under the selected airline brands.

• **Nodes:** airports identified using BTS airport IDs. **Directed edge A → B:** at least one scheduled nonstop flight from A to B during the study window. **Weight:** number of scheduled flights on that ordered route.

• **Network meaning:** scheduled connectivity, not only successfully completed trips. Keep canceled scheduled flights in the construction of the planned network.

• **Comparison:** construct separate AA, DL, and WN networks with identical processing rules. The primary null-model/resilience analysis uses **directed binary topology**; flight-frequency weights support secondary descriptive and strength-based analyses.

5 Existing structural measures and their roles

Degree/strength, degree centralization, strength Gini, betweenness: characterize hub concentration and candidate critical airports. **Clustering, modularity, communities, k-cores, participation/bridging, connectivity:** characterize organization beyond local degree. Degree-based hub concentration is already preserved by the degree-sequence null model; it cannot be claimed as an additional beyond-degree effect.

Use Louvain on a documented undirected projection for community detection, and use a consistent projection for k-core analysis. State explicitly how directed ties and weights are treated. Existing measures should be interpreted in light of H1–H3, rather than presented as an isolated catalogue.

6 Null-model design

For each of the three observed binary directed networks, generate independent randomized networks by directed edge rewiring. The reference ensemble should retain the original airport nodes and preserve **each airport's in-degree and out-degree** exactly, while changing which airports are connected. In a simple directed graph this also preserves the number of edges.

- **Validation:** verify exact degree sequences, node counts, edge counts, and absence of self-loops or duplicate directed edges after rewiring.
- **Randomization quality:** use sufficient valid edge swaps and inspect mixing/variation across independent runs. Do not force strong connectivity, because connectedness is one of the outcomes under investigation.
- **Feasible sampling:** begin with **100** randomized networks per airline; expand toward 200 only if runtime and stability justify it. Use empirical **95% null-model simulation intervals** (2.5th–97.5th percentiles), not confidence intervals for an airline population.
- **Interpretation:** a real-vs-null difference indicates that preserved in/out degrees do not fully account for the result under this model. Since rewiring changes several higher-order properties at once, it does **not** prove that communities or cores caused the difference.

7 Eight-step analysis and implementation plan

Step 1 — Data preparation and validation

Reuse the established BTS-processing pipeline. Validate airport identifiers, route aggregation, flight weights, data coverage, and consistency of network construction across all three airlines.

Step 2 — Characterize hubs and network structure

Calculate existing degree, strength, centralization, strength concentration, betweenness, clustering, connectivity, and reachability measures. Prepare concise descriptive tables to support later interpretations of H1 and H2.

Step 3 — Communities, cores, and bridges

Detect communities with Louvain on a documented undirected projection; compute modularity and k-core structure. Identify high-betweenness and inter-community bridging airports, then assess how these roles relate to degree for H3.

Step 4 — Build degree-preserving baselines

Generate, validate, and store independent directed rewired networks for each airline using the agreed null-model rules. Summarize variation in selected structural properties across the null ensembles.

7 Analysis plan (continued)

Step 5 — Simulate airport removals

Compare **random removal** with removal ordered by **degree** and **betweenness** on original and randomized binary networks. Test approximately **5%, 10%, 15%, and 20%** of airports removed. Fix targeted rankings on each intact network for the primary experiments, recomputing rankings separately for each randomized network. Repeat random selections with recorded seeds; use a manageable number of trials and report their variation. Strength-based targeting on weighted original networks remains supplementary.

Step 6 — Measure structural resilience

Track **directed global efficiency, ordered-pair reachability, and the largest strongly connected component (SCC)** at each removal level. For the primary curves, keep the **original airport set** as the evaluation universe; pairs involving removed or unreachable airports contribute zero. Define efficiency as the mean of 1/d over all distinct ordered pairs (0 for unreachable pairs). Also report efficiency retention relative to each intact network and SCC size relative to original N. Compare original-network outcomes with null distributions.

Step 7 — Estimate airport criticality

Remove each airport from the original networks one at a time; calculate its efficiency loss under the same fixed-node convention. Compare loss rankings with degree, betweenness, strength (supplementary), and existing core/bridging measures. Use Spearman correlations and top-k overlap; for H3, prioritize **degree-matched comparisons** or a degree-only vs. degree-plus-bridging prediction comparison. Full criticality testing on every null-model graph is optional if expensive.

Step 8 — Integrate evidence and report hypotheses

Report findings in the order **H1 → H2 → H3**, reusing the framework on page 2. For each: show the predefined comparison, an effect size or difference, relevant uncertainty/null interval, and whether the evidence supports the prediction. Interpret airline differences cautiously; use structural descriptions to explain, not replace, the experimental results.

8 Analysis conventions to lock before coding

Pre-analysis rule	Agree on the primary removal levels, fixed-ranking protocol, pairwise-efficiency definition, null ensemble size, and evidence criteria for H1–H3 **before inspecting the outcomes**. Keep supplementary analyses labeled as exploratory.

Fair comparisons: apply identical definitions within each airline and its null models, and use normalized outcomes for comparisons across networks of different sizes. Report both relative retention and baseline efficiency where useful; normalization alone can hide differences present before any closure.

9 Four coordinated workstreams

The project remains a single study with four parallel research workstreams. Each student owns a distinct component while using common network definitions, shared data formats, versioned notebooks, and the same hypotheses. Some tasks depend on earlier outputs, so interfaces should be agreed in the first days.

Workstream	Primary responsibility	Shared output
A · Construction	BTS extraction, network validation, degree/strength and base structural characterization.	Three validated network files and reproducible summary tables.
B · Structure and nulls	Louvain, modularity, k-cores and bridging roles; directed degree-preserving rewiring.	Community labels, airport measures, validated null ensembles.
C · Resilience	Random/targeted removal experiments, fixed-universe resilience metrics and curves.	Matched real-vs-null resilience results for H1/H2.
D · Criticality and synthesis	Single-airport removal, degree-controlled comparisons, H3 evaluation, coordinated interpretation.	Criticality rankings, hypothesis evidence table, integrated results.

10 Feasible three-week sequence

Week 1: Lock definitions and the hypothesis-to-analysis framework; validate networks; implement community measures and a small null-model pilot. **Week 2:** Scale and validate randomized ensembles; run resilience experiments and single-airport criticality; check consistency. **Week 3:** Complete degree-controlled comparisons, validate figures, interpret H1–H3, and write/review the final report.

11 Limitations and interpretation rules

- This is a **structural** resilience study. It does not measure passenger volumes, capacities, rerouting feasibility, layovers, or real-world operational losses. Airport closures are hypothetical.
- Degree-preserving rewiring controls degree sequences, but not weights or all possible constraints of airline operations. Communities and criticality measures are observational structural correlates, not proven causal mechanisms.
- With just three airlines, comparisons are **case studies**, not a statistical inference about all airlines. Random airport removal is a baseline, not a realistic probability model for disruptions.

12 Expected deliverables

Three documented airline networks; reproducible Python notebooks; descriptive/community figures; validated directed null ensembles; resilience curves with null-model intervals; airport criticality rankings with degree-controlled evidence; and a final PDF report (up to 5,000 words) organized around H1–H3.

What makes it strong	The project tests an integrated explanation: **what degrees already account for, what the observed arrangement of connections adds, and which airports matter most**—with explicit comparisons and clear limits on interpretation.